**CPS 4745-01/Fall 2026** **Visualization Design and Development** **Dr. Huang**

**Instructor**: Dr. Ching-yu (Austin) Huang **Website**: http://austinhuang.net/ **Email**: chuang@kean.edu
**Office Hours**: https://yoda.kean.edu/chuang Mon, Tue, Wed, Thur: 10:00–11:15 am
**Class**: **Monday**, **Wednesday** (**11:15 - 12:30 pm**), **Location: GLAB 207**
**Prerequisites**: **CPS 3740** and **MATH 2526**. **Students without prerequisite(s) must withdraw from the class.**

| WK | | Date | Topic | Notes | # |
|---|---|---|---|---|---|
| 1 | 9/2/26 | Wed | Overview + Visualization intro, online visualization | DB account, tools, env | 1 |
| 2 | 9/7/26 | Mon | **Labor Day** | **No class** | |
| | 9/9/26 | Wed | XAMPP, MySQL Workbench, basic SQL filters | **Study 1 starts** | 2 |
| 3 | 9/14/26 | Mon | Visualization and charts using Excel -1, storytelling | | 3 |
| | 9/16/26 | Wed | Basic SQL Aggregations | | 4 |
| 4 | 9/21/26 | Mon | Trendline, correlation & outliers in Excel | | 5 |
| | 9/23/26 | Wed | Visualization and charts using Excel -2, storytelling | | 6 |
| 5 | 9/28/26 | Mon | DV goal and SQL CREATE, data marts | | 7 |
| | 9/30/26 | Wed | Data Visualization in Google Looker, import dataset | | 8 |
| 6 | 10/5/26 | Mon | Study 1 presentation – DV goals, tools, data marts in Excel | Study1 due (10/4) | 9 |
| | 10/7/26 | Wed | Study 1 presentation – Continue | | 10 |
| 7 | 10/12/26 | Mon | Midterm exam (day 1 – 10) | **Study 2 starts** | 11 |
| | 10/14/26 | Wed | Identify your datasets, data marts, storytelling | | 12 |
| 8 | 10/19/26 | Mon | Load data into the database, create data marts in SQL | | 13 |
| | 10/21/26 | Wed | SQL for pivot, merge data marts, SQL JOIN | | 14 |
| 9 | 10/26/26 | Mon | Normalization and binning for data marts and visualization | | 15 |
| | 10/28/26 | Wed | Web technology, client server, 3-tier, HTML, CSS, JS | | 16 |
| 10 | 11/2/26 | Mon | HTML, , Elements, JavaScript DOM Access, JSON | | 17 |
| | 11/4/26 | Wed | HTML FORM, inputs, JavaScript debugging | | 18 |
| 11 | 11/9/26 | Mon | Study 2 presentation – visualize your datasets in Looker | Study2 due (10/4) | 19 |
| | 11/11/26 | Wed | Study 2 presentation – Continue | | 20 |
| 12 | 11/16/26 | Mon | CGI, PHP MySQL | **Team project starts** | 21 |
| | 11/16/26 | Mon | **Last day to withdraw with 'W' grade** | 0% refund | |
| | 11/18/26 | Wed | Google Charts | | 22 |
| 13 | 11/23/26 | Mon | Dynamic data visualization with PHP MySQL | | 23 |
| | 11/25/26 | Wed | JavaScript events & handling | | 24 |
| 14 | 11/30/26 | Mon | Interactive data visualization | | 25 |
| | 12/2/26 | Wed | Continue, web hosting | | 26 |
| 15 | 12/7/26 | Mon | Final exam (day 1 – 26) | Final exam | 27 |
| | 12/9/26 | Wed | **School follow Tuesday schedule** | No class | |
| 16 | 12/14/26 | Mon | Team project presentation | Team proj. due (12/14) | 28 |
| | 12/16/26 | Wed | Team project presentation – Continue | | 29 |
| 17 | 12/21/26 | Mon | Feedback, storytelling and insights (remote) | | 30 |

**Late revision: 10/23/2026. For important dates, please consult the Academic Calendar via the following link:**
**https://www.kean.edu/offices/registrars-office/academic-calendar**

**Content**: This course will focus on software design and development to represent data in visual formats from various data sources. Students will learn how to develop applications and present information from the database in an understandable and effective methods in charts, graphs, dashboards and dynamic web pages. Students will do hands-on projects and compare their results with popular visualization software.

**Course Student Learning Outcomes: Upon completion of this course, the student will be able to:**

A. Explain the key techniques and theory used in visualization, including data models, graphical perception and techniques for visual encoding and interaction.
B. Present information in an understandable, effective, and aesthetic manner, for the purposes of explaining ideas and analyzing data.
C. Utilize popular API and software to visualize data
D. Design visual tools that allows user to visualize and interact with data from the database.

**Instructional Method**

- Slides, assignments, and tool info will be posted on the class google drive that will be sent by the instructor.
- For more information about this course and for course materials, go to https://kean.instructure.com/

**Textbook & Materials**

Textbook is required. Refer to the slides and the URLs in the slides.
Interactive Data Visualization for the Web, 2nd Edition.
Author: Scott Murray
ISBN-10: 9781491921289, ISBN-13: 978-1491921289
Publisher: O'Reilly Press, 2017. *(Online)* Code examples available on GitHub.

**Require hardware and Software**

- This is a hands-on course. Students are required to bring a personal laptop to every class.
- Database related free software: MySQL Workbench, XAMPP, MySQL
- Other free software: Putty.exe (Win), terminal (Mac), FileZilla, Sublime
- An account on a Linux server will be provided by the instructor.

**Open Educational Resources**

For more information, please see the Nancy Thompson Library Learning Commons:
http://libguides.kean.edu/OER

**Assessment and course grading**

The course includes Exams, Labs, Homework and Project, a **total of 1000 points**.

- 1 Midterm and 1 final exam (225, 225 pts)
- Individual Study 1 and presentation (150 pts) using Google Sheets or MS Excel
- Individual Study 2 and presentation (150 pts) using DV tools – Tableau, Google Looker, MS Power BI.
- Team Project and presentation (150 pts) using JavaScript, Ajax, CSS, HTML, PHP, MySQL
- Attendance (100 pts, 4pts/each class, must do exercises and attend at least 60 minutes/each class)

**The Digital Gradebook (in Canvas) will be used to record student grades.**

**Grading Policy**

| | | |
|---|---|---|
| **A: >= 940** | **A-: 939 -890** | |
| **B+: 889 - 840** | **B: 839 - 800** | **B-: 799 – 760** |
| **C+: 759 – 720** | **C: 719 - 680** | |
| **D: 679 – 640** | | |
| **F: below 640** | | |

**C or better is needed for CS and IT majors.**

**Technical Requirements:**

1. In order for your Canvas course to function correctly, you will need to disable pop-ups on your Internet browser.
2. Make sure you have Microsoft Office installed on your computer. You may be eligible for a free MS Office Software Student Edition. You are required to create an account and provide a valid Kean University ID to obtain access to the software applications. To start the application process, go to the Office 365 Education website.

**Academic Early Alert** information will be provided to students by the instructor via KeanWise. Please see the 2020-2021 Undergraduate or Graduate Catalog for more information.

**Class Policies:**

- Posting of course materials: **Public posting of course materials from this class is prohibited.** Materials presented in the class are for the personal use of registered students during the semester only. Homework, programs, and other classwork should not be shared, distributed or publicized in any way, for the protection of other students and in respect of the intellectual property rights of the faculty and copyright owners. Kean University's Academic Integrity Policy can be found here: https://www.kean.edu/media/academic-integrity-policy.
- No late work will be accepted without permission in advance from the instructor. Late work is penalized.
- No make-up exams are provided without documentation of medical or family emergency. Please arrange your schedule for the next 16 weeks to arrive in class on time and prepared.
- Attendance and participation are required.
- All homework assignments, projects, and exams are expected to be completed by the students themselves. **No credit will be given for an assignment or homework that is copied – in part or in total – from another person. The use of AI tools such as ChatGPT, GitHub Copilot, and similar resources is not permitted for these tasks**.
- Exam policy: During class examinations, if a student leaves the classroom for any reason during the exam time, **the exam must be turned in before the student departs, concluding the exam for that student**. The student cannot resume the exam on return later. You may not make or take a phone call, text or access the internet during class or a test.
- **Classes may not be audio or digitally captured**, unless permission has been granted by the instructor for that specific class.

For remote learning only - Class Recordings (Live Sessions/Collaborate)

In this class software may be used to record live class discussions. As a student in this class, your participation in live class discussions will be recorded. These recordings will be made available only to students enrolled in the class, to assist those who cannot attend the live session or to serve as a resource for those who would like to review content that was presented. All recordings will become unavailable to students in the class when the course is closed.

Students are not permitted to record (or download) the course experience. If a recording of the course is required for accommodation purposes students must make those arrangements with the Office of Accessibility Services. Students found to have recorded (or downloaded) course experiences without authorization are subject to review by the Office of Student Conduct and Community Standards.

Privacy Statement

**Assuring privacy among faculty and students engaged in online instructional activities helps promote open and robust conversations and mitigates concerns that comments made within the context of the class will be shared beyond the classroom. As such, recordings of instructional activities occurring in online classes may be used solely for internal class purposes by the faculty member and students registered for the course, and only during the period in which the course is offered.**

## Important University Policies and Information

- All students and faculty should familiarize themselves with the **Cougar Restart Guide** which includes important virtual resources for students by Kean University via the following link: https://www.kean.edu/coronavirus-information/resources-current-prospective-students
- Students are responsible to review and understand the *University Academic Integrity Policy* available via the following link: https://www.kean.edu/academic-integrity
- *Instructors must indicate on the syllabus and/or individual assignment if the use of Kean University academic support services (Tutoring and/or Writing Center) is permitted for take home assignments/exams.*
- Students should review the *Student Code of Conduct*, as it discusses expectations of appropriate conduct in the classroom: https://www.kean.edu/offices/community-standards-and-student-conduct/student-code-conduct
- Students are strongly encouraged to register for the University's emergency notification system (http://www.kean.edu/campusalert) in order to be informed of campus emergencies, weather notices, and other announcements. Kean Ocean students would sign up via the following link: https://ocean.sendwordnow.com/LicensePage.aspx
- All students must have a valid Kean email account. For those who do not already have one, forms are available on-line at https://www.kean.edu/offices/computer-and-information-services/ocis-forms click on E-mail Account Request Form.

## Diversity & Non-Discrimination Statement

Kean University is committed to establishing and maintaining a diverse campus community through inclusive excellence and equal opportunity. Kean's commitment to access and equity is designed to prepare each graduate to not only thrive, but climb higher in a diverse world. As an affirmative action, equal opportunity institution we work to support a campus-wide agenda to foster a community that both values and promotes the diversity and equity of all students, faculty, staff, administrators, and beyond. Kean adheres to the *New Jersey State Policy Prohibiting Discrimination in the Workplace;* for more information on Kean's non-discrimination policy and procedures, and protected categories under the policy, please visit: https://www.kean.edu/offices/affirmative-action.

## Americans with Disabilities Statement

Students with documented disabilities who may need instructional accommodations or special arrangements in the event of an evacuation should notify the instructor as soon as possible and no later than the second week of the term. Students may also contact The Office of Accessibility Services by phone at (908) 291-3137 or email accessibilityservices@kean.edu to discuss special needs, accommodations and arrangements.

## Title IX

Title IX of the Education Amendments of 1972 (Title IX) prohibits discrimination on the basis of sex in education programs or activities. Sexual harassment in any form will not be tolerated at Kean University. Sexual harassment by students should be reported to the Office of Affirmative Action Programs, Office of the Vice President for Student Affairs or the Office of Community Standards and Student Conduct immediately. Information about the University's Sexual Misconduct Policy may be found at the following: https://www.kean.edu/offices/policies/sexual-misconduct-policy.

## Syllabus Updates:

**Instructor reserve the right to change the syllabus at any time throughout the semester.** All changes will be announced in class and a new syllabus will be electronically available to all students. If you are enrolled in this class past the add/drop date you are subjected to all rules in this syllabus.